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INTRODUCTION TO CRYPTOGRAPHY

AN OVERVIEW OF CRYPTOGRAPHY

- ✕ The practice of securing communication so that only the intended audience can read or process the information.
- ✕ It's no longer just about hidden letters; it's the backbone of the entire internet—protecting your passwords, bank transfers, and private chats.
- ✕ Modern cryptography exists at the intersection of the disciplines of mathematics, computer science, electrical engineering, communication science, and physics.
- ✕ Teaches how to convert sensible data into random junk!



1. PRIVACY / CONFIDENTIALITY

Ensures that **no one** can read the message except the **intended receiver**.



2. INTEGRITY

Assures the receiver that the message has **not been altered** in any way from the original.



3. AUTHENTICATION

The process of **proving one's identity**. Are you who you say you are?



4. NON-REPUDIATION

A mechanism to prove that the **sender really sent** the message and **cannot deny it** later.



5. KEY EXCHANGE

The method by which **crypto keys** are **shared securely** between sender and receiver to start communication.





WITHOUT CRYPTOGRAPHY (INSECURE)

Messages travel in **plain text**, anyone can **read** or **steal** the data.



Anyone on the network can see or change the message.



WITH CRYPTOGRAPHY (SECURE)

The message is **encrypted** before sending. Only the intended **receiver** can **decrypt** and read it.





CRYPTOGRAPHIC TERMS



PLAINTEXT

- ▶ **Original** message
- ▶ **Anyone** can **read**
- ▶ E.g., "Alice"



CIPHER

- ▶ Cryptographic **algorithm** or **function**
- ▶ Tells **how to transform** plaintext
- ▶ into that random junk



CRYPTOGRAPHY CONCEPTS

THE LANGUAGE OF SECURE COMMUNICATION



CRYPTOGRAPHIC TERMS



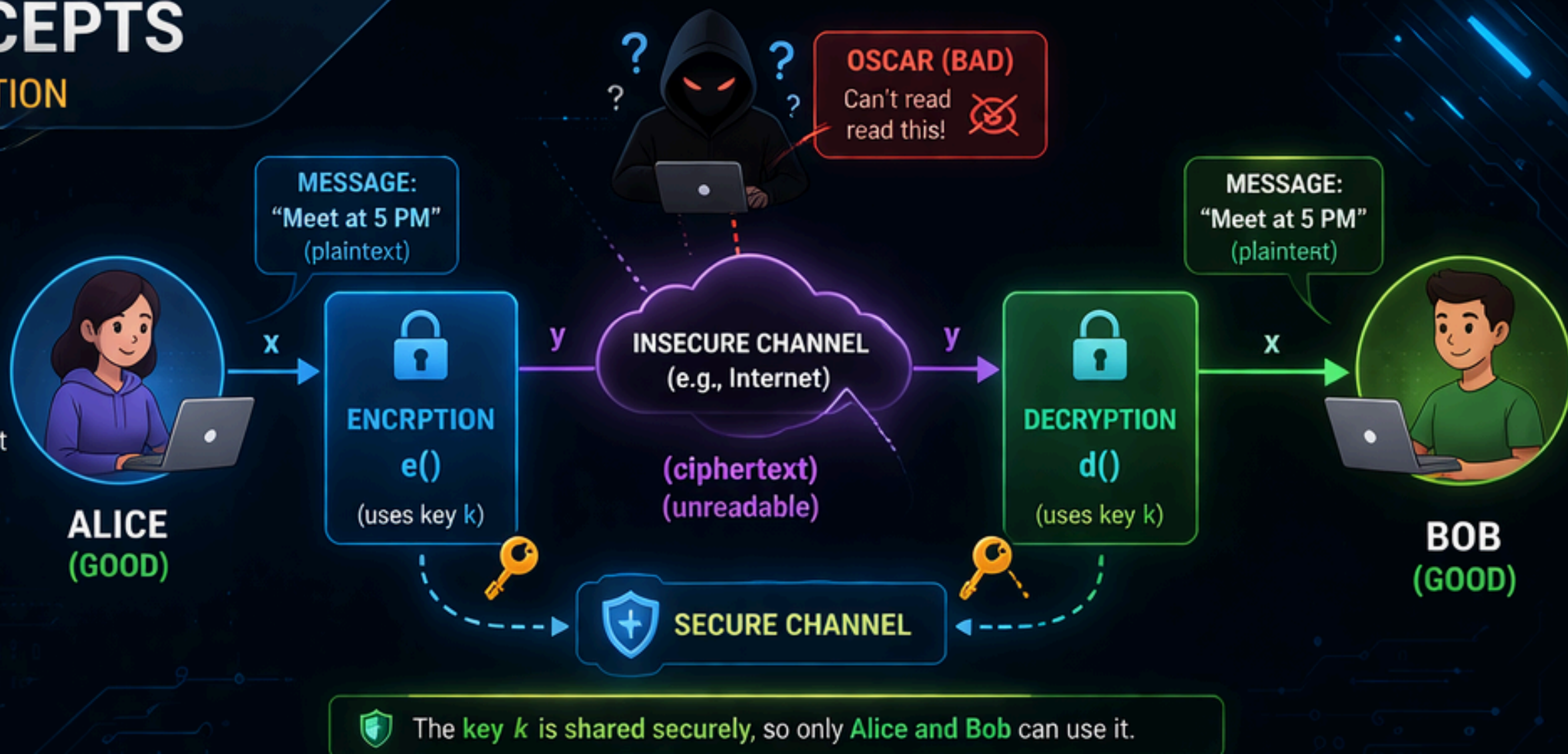
CIPHER KEY

- ▶ **string** of characters
- ▶ Cipherfunction(plaintext + cipher key) = ciphertext



CIPHER TEXT

- ▶ That **random junk** we got after applying cipher key on plaintext
- ▶ Unreadable, **useless**
- ▶ E.g., "#r4Tf2%#"



CRYPTOGRAPHY CONCEPTS

THE ART OF SECURE COMMUNICATION

CRYPTOGRAPHIC TERMS

ENCRYPTION

Converting **plaintext** into **ciphertext** using **cipher key**.

Cannot be reversed without the use of the **key**.



Scrambles readable data into **unreadable** format.

DECRYPTION

Converting **ciphertext** into **plaintext** using **cipher key**.

Cannot be reversed without the use of the **key**.



Restores unreadable data back to its **original** meaning.



Encryption protects data. Decryption reveals it.
THE KEY IS EVERYTHING!

CRYPTOGRAPHY CONCEPTS

BUILDING BLOCKS OF DATA SECURITY

CRYPTOGRAPHIC TERMS

HASH

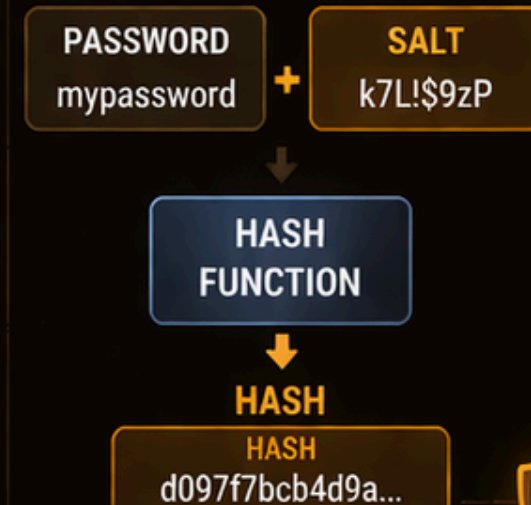
- ✓ Random **fixed length** string
- ✓ **Irreversible one way** function
- ✓ Provides **integrity**
- ✓ E.g., **MD5, SHA-1**



Same input = Same hash
(Tiny change = Completely different hash)

SALT

- ✓ Additional **security**
- ✓ Eradicates **brute force** and **cracking**
- ✓ Random string **appended** at beginning or end of plaintext



HASH ensures integrity. SALT ensures security.
TOGETHER, THEY KEEP ATTACKERS OUT!





CRYPTOGRAPHY PRACTICAL TIME!

See it. Try it. Understand it.



BEST WEBSITES TO EXPLORE CRYPTO



1. CRYPTOOOL

<https://cryptool.org/en/cto/>

Learn & experiment with algorithms (Caesar, AES, RSA)

Algorithms		
Caesar	Classic cipher	
AES	Symmetric	
RSA	Asymmetric	



2. BOXENTRIQ

<https://www.boxentriq.com/code-breaking>

Try classic ciphers and see how they work

Caesar Cipher		
Input		
HELLO	3	3
Shift	3	
KHOOR		Output



3. MDSONLINE

<https://www.md5online.org>

Generate MD5, SHA-1, SHA-256 hashes instantly

Text		
Hello Alice!		
MD5	SHA-1	SHA-256
64489c85dc2fe0787b85...		



4. ASYMMETRIC ENCRYPTION DEMO

<https://travistidwell.com/jsencrypt/>

Try RSA encryption in your browser

Public Key	
-----BEGIN-----	
Message	
Hello World!	Encrypt



5. CYBERCHEF (BY GCHQ)

<https://gchq.github.io/CyberChef/>

A hacker's Swiss Army knife for encoding, decoding, hashing & more

FROM	TO
UTF-8	Hex
Hello	= 48656cc6f



TIP: Open these sites, try it live, and show the magic to your class!



PRACTICAL DEMO IDEAS

(LIVE IN CLASS)

Pick one and do it together!

1. CAESAR CIPHER (ENCRYPTION/DECRYPTION)

Website: boxentriq.com/code-breaking

What to do: Select **Caesar Cipher**, set **shift = 3**

Encrypt "ATTACK" → "DWWDFN"

Then decrypt it back!

CAESAR CIPHER

Action	Encrypt
Shift	3
Input	ATTACK
Output	DWWDFN

Try changing the shift!

2. HASHING DEMO (MD5 / SHA-256)

Website: md5online.org

What to do: Type any text like "Hello Alice!"

Click **MD5** and **SHA-256**

Show how even 1 letter change = completely different hash!

Text
Hello Alice!

MD5

64489c85dc2fe0787b85c87214b3810

SHA-256

185f8db32271fe25f561a6fc938b2e2b4
306ec30deda518007d1764f826381969

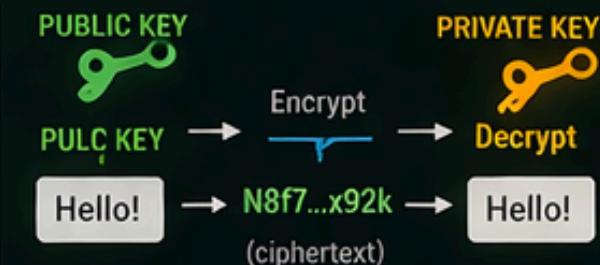
Change one letter and see!

3. RSA ENCRYPTION (PUBLIC KEY)

Website: travistidwell.com/jsencrypt/

What to do: Generate keys, type a message, encrypt it, then decrypt it.

Show how public key encrypts but only private key decrypts!



Magic of asymmetric crypto!

4. CYBERCHEF CHALLENGE (FUN!)

Website: gchq.github.io/CyberChef/

What to do: Convert "Hello" to Hex, Base64, URL Encode, and back.

Let students try decoding something mysterious!

INPUT	RECIPE	OUTPUT
Hello	From From-5 UTF-8	48656c6c6f
	To Hex	

Try different recipes!



CLASS ACTIVITY IDEA

Split students into small groups and give them challenges:



Challenge 1

Decrypt this Caesar cipher text: **KHOOR** (shift = 3)



Challenge 2

Generate SHA-256 hash of your name



Challenge 3

Encrypt a message using RSA and share the ciphertext



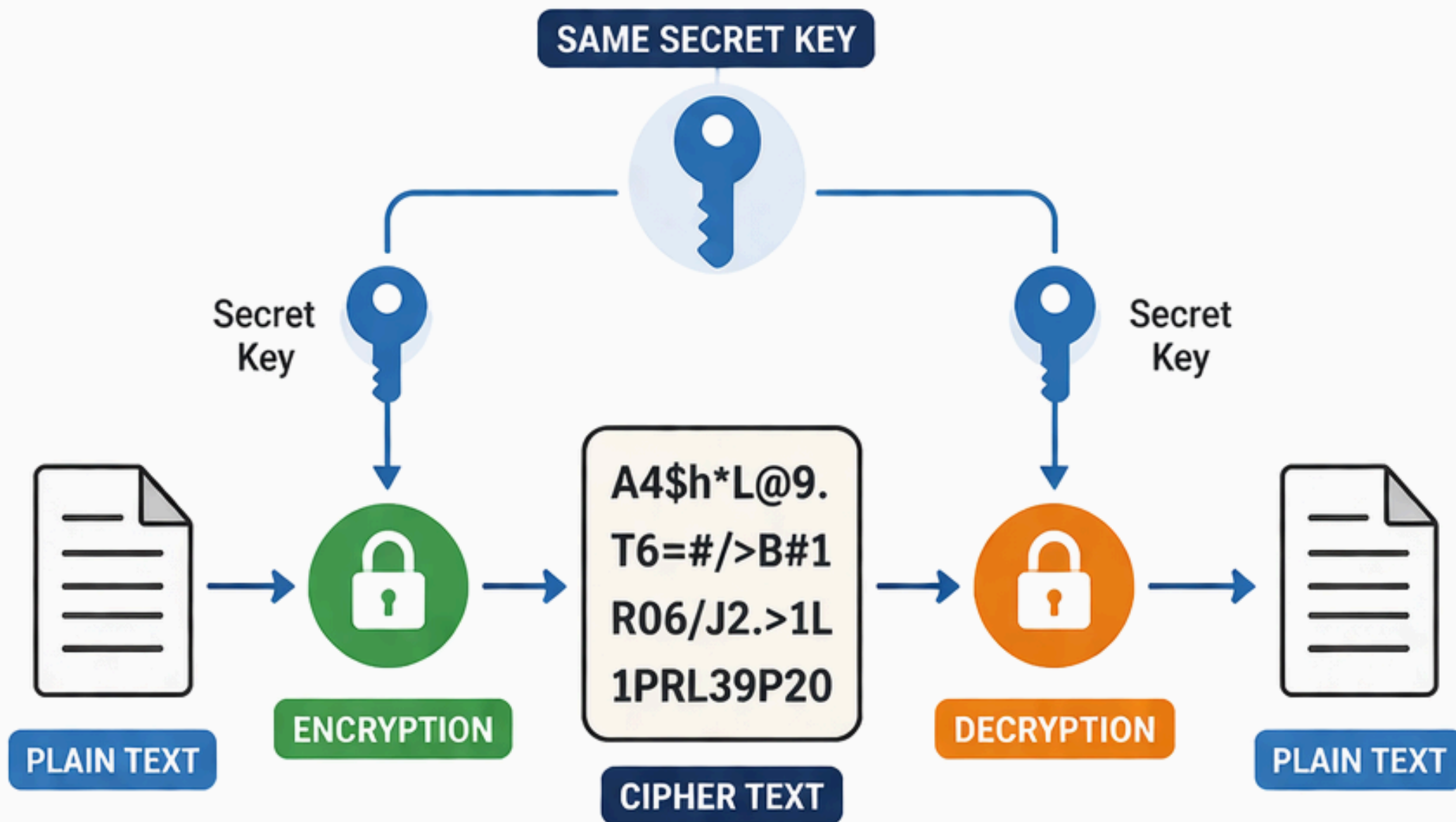


Cryptography Concepts

Understanding Symmetric or Secret Key Cryptography

SYMMETRIC (SECRET KEY) CRYPTOGRAPHY

The **same secret key** is used for both encryption and decryption.



Plain Text → Encryption (with Secret Key) → Cipher Text
Cipher Text → Decryption (with **Same Secret Key**) → Plain Text

✓ ADVANTAGES

- ✓ **Extremely Secure**
Strong encryption makes it hard for attackers to break.
- ✓ **Relatively Fast**
Uses less computational power, so it's efficient.



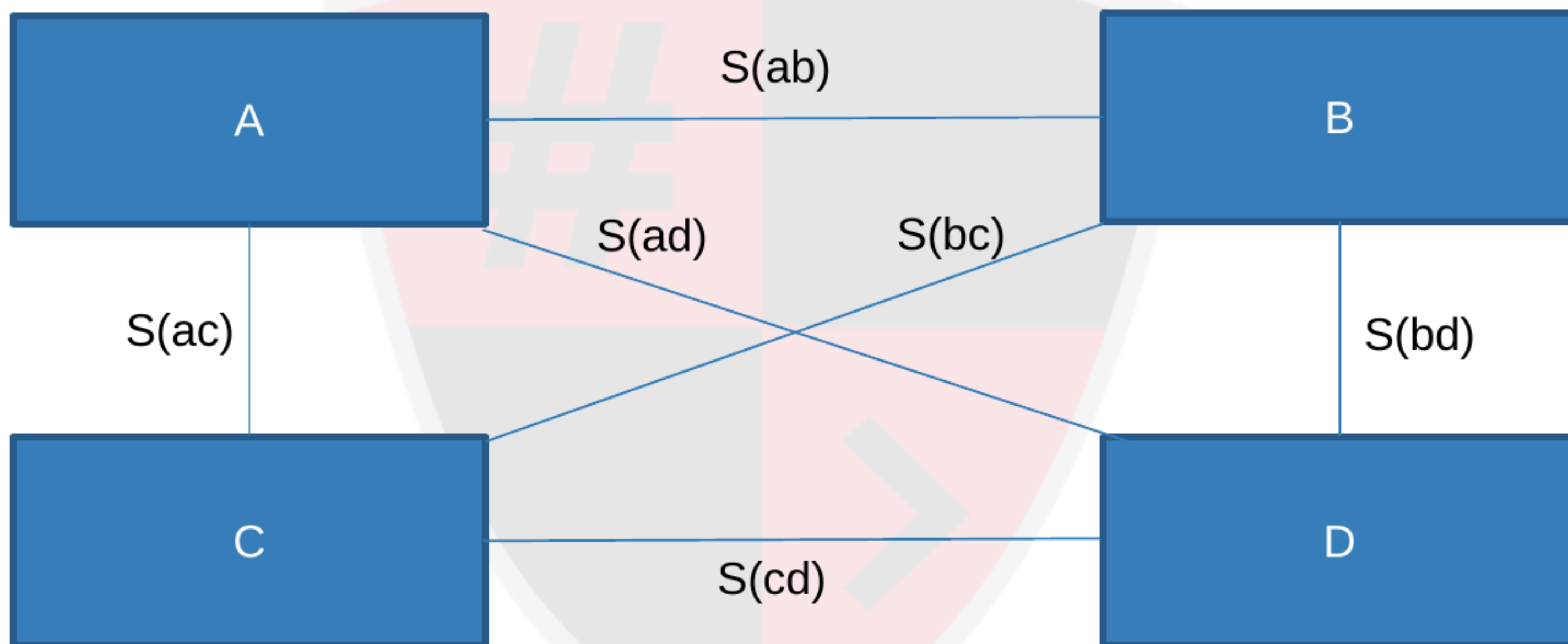
✗ DISADVANTAGES

- ✗ **Key Management**
Storing and managing keys securely is challenging.
- ✗ **Key Distribution**
Sharing keys between parties safely is difficult.
- ✗ **Large Number of Keys Needed**
For n users, we need $O(n^2)$ keys.



Cryptography concepts

■ Symmetric or Secret Key Cryptography





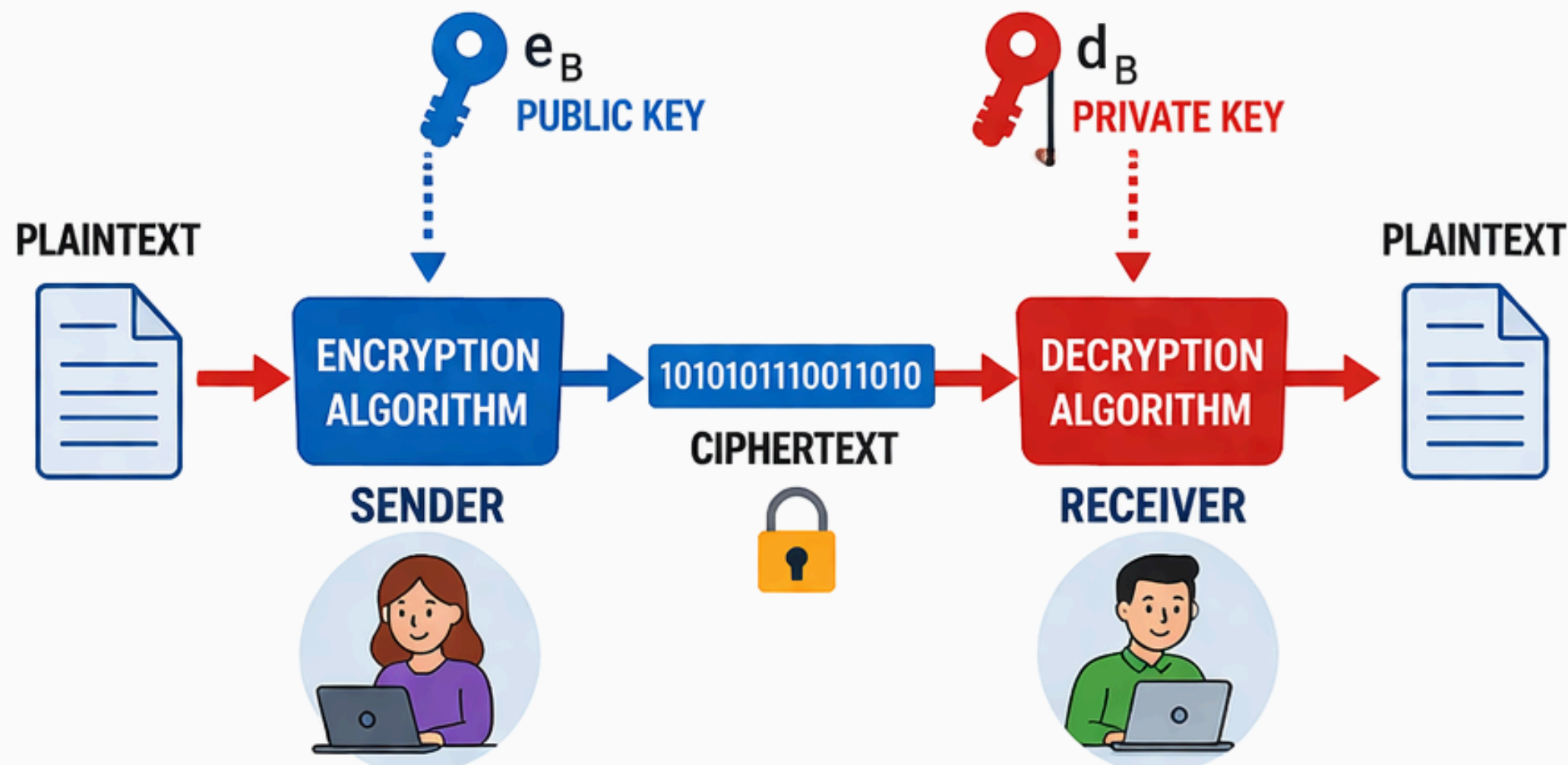
CRYPTOGRAPHY CONCEPTS

Understanding Asymmetric or Public Key Cryptography

ASYMMETRIC (PUBLIC KEY) CRYPTOGRAPHY

Uses a pair of keys: a **public key** for encryption and a **private key** for decryption.

 e_B Public Encryption Key
 d_B Private Decryption Key



✓ ADVANTAGES

-  **Easier Key Management**
No need to share private keys.
-  **Easier Key Distribution**
Public keys can be shared openly.
-  **Less Number of Keys Needed**
For n users, we need $O(n)$ keys.

✗ DISADVANTAGES

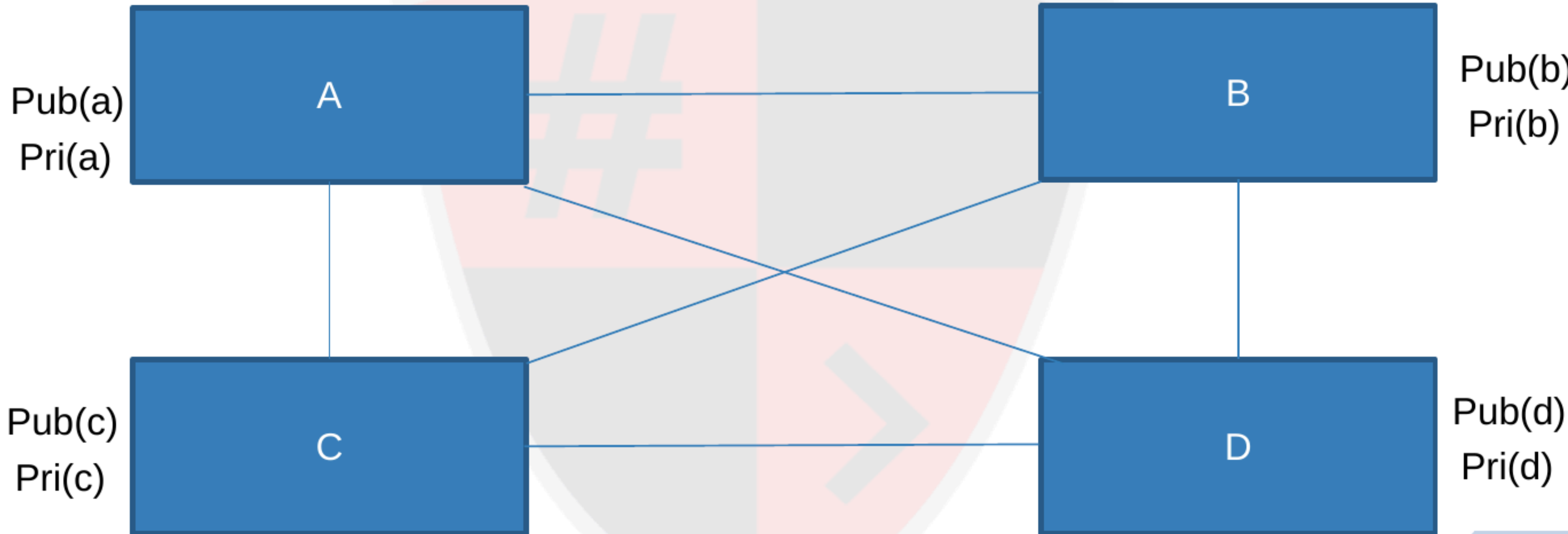
-  **Relatively Slower and Complex**
Requires more computational power.
-  **Management of Public Keys**
Verifying and managing public keys can be challenging.

Plaintext → Encryption (with **Public Key**) → Ciphertext → Decryption (with **Private Key**) → Plaintext



Cryptography concepts

Asymmetric or Public Key Cryptography



CRYPTOLOGY

The science of securing information by transforming it to make it unintelligible to unauthorized users.

